

B.Tech RND Project

Low-Light Image Enhancement

Yacha Venkata Sai Hitesh

Department of CSE, IIT Dharwad

Under the Guidance of :

Professor Dileep A.D

Professor Vandhana Bharti

Outline

1. Introduction & Motivation
2. Challenges — What Goes Wrong in Low-Light Images
3. Baseline Model — CIDNet and the HVI Colour Space
4. Five Identified Limitations of CIDNet
5. Innovation Category 1 — Noise-Adaptive Processing
6. Innovation Category 2 — Frequency-Based Processing
7. Innovation Category 3 — Attention-Adaptive Processing
8. Full Architecture Overview
9. Experimental Setup
10. Results and Comparison
11. Ablation Study
12. Novelty and Contributions
13. Conclusion and Future Work

Introduction — Why Low-Light Enhancement Matters

Real-world applications critically depend on image quality in dark environments:

- **Security & Surveillance** — most criminal incidents occur at night; cameras must identify faces, plates, behaviour
- **Autonomous Vehicles** — pedestrian/obstacle detection in darkness directly impacts road safety
- **Medical Imaging** — endoscopic cameras in dark body cavities; clearer images improve surgical decisions
- **Wildlife Research** — camera traps for nocturnal animals produce noisy, underexposed images
- **Smartphone Photography** — Night Mode computational pipelines used by billions of users daily

Goal: Given a degraded low-light image I_{low} correct the brightness, noise, colour, texture and produce the enhanced image.

Challenges — What Happens in Low-Light Images

Four mainly challenging problems which makes LLIE really hard are:

1. Noise (Gaussian + Poisson)

Gaussian: electronic readout noise (constant). Poisson: photon shot noise (signal-dependent). Both are present simultaneously and vary spatially.

2. Colour Collapse in Darkness

RGB channels approach zero in dark regions — colour information becomes indistinguishable from noise. Recovering accurate hue from near-zero values is mathematically ill-posed.

3. Brightness-Noise Coupling

Dark $\neq$ Noisy. A dark image from a clean sensor may be noise-free; a moderately bright ISO-6400 image may be severely corrupted. Models that assume dark=noisy fail on clean dark images.

4. Loss of Fine Detail

Edges, textures, and high-frequency structural information are the first casualties of low-light degradation. Pixel-domain losses (L1, SSIM) do not strongly penalise their absence.

Baseline Model — CIDNet and the HVI Colour Space

CIDNet (2024) — starting point, achieving 22.88 dB PSNR on LOLv1.

The HVI Transform

HVI decouples colour from brightness into three channels:

- H and V — encode hue as cosine and sine projections
- I — encodes intensity (brightness)

The collapse factor k controls how aggressively colour is compressed in dark regions:

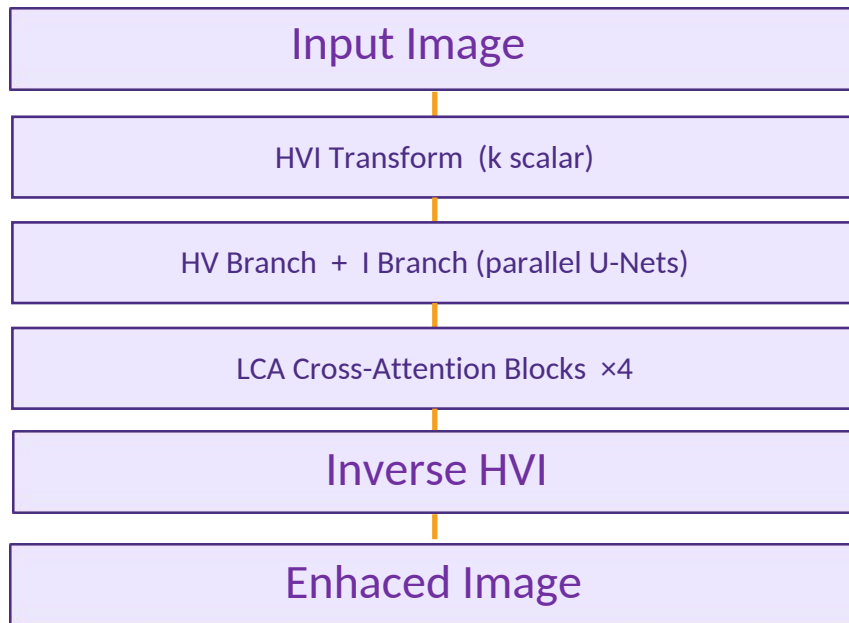
$$cs = (\sin(I \cdot \pi/2) + \epsilon)^k$$

$$H = cs \cdot sat \cdot \cos(2\pi h)$$

$$V = cs \cdot sat \cdot \sin(2\pi h)$$

When $I \rightarrow 0$ (dark), $cs \rightarrow 0$, collapsing H and V toward zero — encoding noise-dominated colour as near-zero.

Processing Pipeline



Five Identified Limitations of CIDNet

After Analysis the research paper and the code I identified five fundamental limitations they are:

L1

Brightness-Only Collapse

k is a single global scalar — same for every pixel, every image. Noise depends on ISO/sensor temperature, not brightness. Dark $\neq$ Noisy.

L2

No Frequency Separation

Noise is high-frequency. CIDNet applies HVI directly to raw input, coupling structural colour with noise-induced fluctuations in one undifferentiated step.

L3

Fixed 3×3 Cross-Attention

K/V sampling restricted to a fixed 3×3 window. Relevant context (displaced bright patch, clean edge) may lie beyond this neighbourhood — unreachable.

L4

Naive Skip Connections

Hard concatenation blindly merges noisy encoder features into decoder — pulling reconstruction toward the degraded input in dark regions.

L5

Incomplete Loss Supervision

No frequency-domain penalty, no differential chroma weighting (H,V carry all colour fidelity), no direct training signal to noise estimation.

Innovation Category 1 — Noise-Adaptive Processing

Addresses L1 and L5

The base model assumes that dark pixels are noisy and it reduces the colour information, it uses fixed k for all pixels, so I implemented simple CNN model to capture noise and based on this noise I updated the k value.

Three innovations to solve the above problem:

- 1. NA-HVI Transform — Noise-Adaptive HVI collapse factor**
- 2. Noise Map Consistency Loss — self-supervised noise estimator training**
- 3. Noise-Level Augmentation — broader noise distribution during training**

Key Insight

Original CIDNet uses a single scalar k — the same collapse for a clean dark image and a severely noisy one. this adaptive k_{map} distinguishes between these cases using a learned per-pixel noise confidence map.

$$k_{\text{map}} = \text{base_k} + \text{noise_gain} \times \text{noise_map} - \text{brightness_gain} \times \text{intensity}$$

Dark + Noisy $\rightarrow$ large k_{map} $\rightarrow$ strong collapse.

Dark + Clean $\rightarrow$ less k_{map} $\rightarrow$ gentle collapse.

Test images used



These images are used to show the below differences.

Results of Noise Adaptive Processing



Based Model



Implemented Model

If you observe region near lamp or right corner of wall in the base model there is light red colour which is because the model assumes dark pixel as noise even if it is clean.

Innovation Category 2 — Frequency-Based Processing

Addresses L2 and L5

We need to separate structure details of objects ,etc in image before enhancing image or else we might loose some details while suppressing noise.

Innovations:

1. **FG-HVI Transform — frequency-guided decomposition before HVI**
2. **Learnable Hue Rotation — correct global colour cast before projection**

FG-HVI Transform — Pipeline:

1. Learnable 15×15 Gaussian blur kernel ($\sigma_0=3.0$, normalised after each update) $\rightarrow$ low_freq component
2. $\text{high_freq} = \text{input} - \text{low_freq}$ (captures edges, texture, noise)
3. Frequency gate CNN predicts $\alpha \in [0,1]$ per pixel from $[\text{input}, \text{low_freq}, \text{high_freq}]$ (9 channels)
4. $\text{blend} = \alpha \times \text{low_freq} + (1-\alpha) \times \text{input} \rightarrow$ only this blend enters the HVI transform
5. After inverse HVI: $\text{output} = \text{enhanced_rgb} + (1-\alpha) \times \text{high_freq} \rightarrow$ reinjects genuine texture

Results of Frequency based processing



As you can see, implemented model was able to get sharp edges of watch, but base model failed to retrieve the details and also the gray spots are due to base model assuming dark pixel as noise (explained before).

Innovation Category 3 — Attention-Adaptive Processing

Addresses L3, L4, L5

Learn where to focus processing capacity and which information sources to trust.

Six improvements in this category:

Dynamic Skip Gate

- 1 Replaces hard skip concat with a learned sigmoid gate — suppresses noisy encoder features in dark regions.

Deformable Cross-Attention

- 2 Replaces fixed 3×3 attention with deformable K/V sampling. Deformable Cross-Attention is used to capture distant(far) information

Channel, Spatial Attention

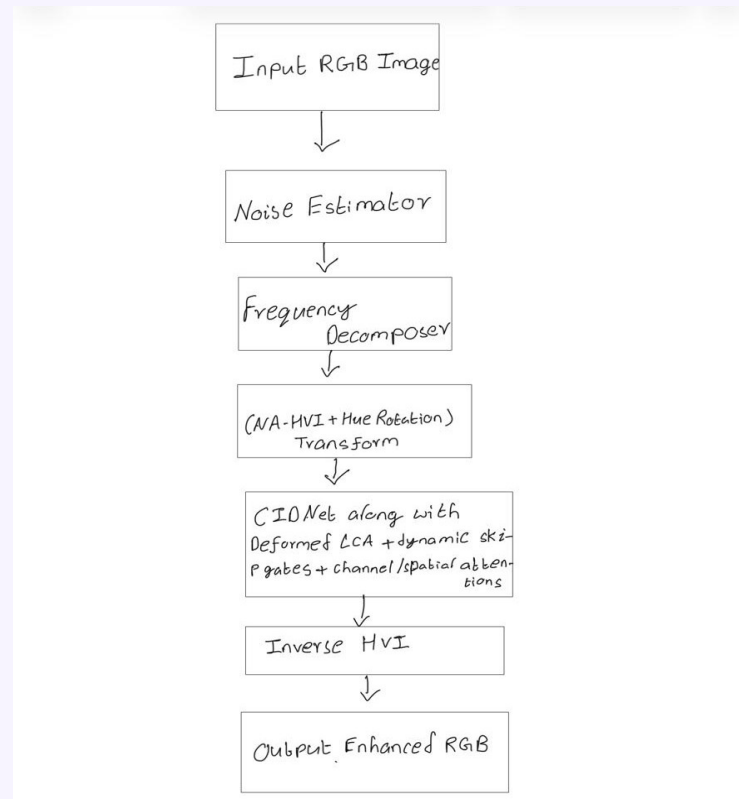
- 3 Channel attention selects *what features* are important by emphasizing useful channels and suppressing noisy ones, while spatial attention decides *where to focus* by prioritizing important regions for stronger enhancement.

Full Architecture Overview

Complete forward pass through the enhanced model:

1. NoiseEstimator CNN $\rightarrow$ per-pixel noise map $\hat{n}(x) \in [0,1]$
2. FrequencyDecomposer $\rightarrow$ low_freq, high_freq, gated blend image, gate α
3. NA-HVI Transform (blend + $\hat{n}$) $\rightarrow$ HVI tensor with adaptive k_map
4. Dual-Branch U-Net (HV branch + I branch with Deformable LCA + Dynamic Skip Gates + Channel/Spatial Attention)
5. Inverse HVI + high-frequency reinjection $\rightarrow$ final enhanced RGB output

Training loss = Original (L1 + SSIM + Edge + Perceptual) + New (FFT + Chroma + Noise Consistency)



Experimental Setup

Dataset — LOLv1

- 485 training pairs, 15 test pairs (eval15)
- Real-world low-light / normal-light aligned pairs
- Diverse indoor & outdoor scenes, $\sim 400 \times 600$ px
- Captured by varying camera exposure settings

Evaluation Metrics

- PSNR (dB, $\uparrow$) — pixel-level fidelity; each +1 dB is meaningful
- SSIM ([0,1], $\uparrow$) — structural and textural similarity
- LPIPS ($\downarrow$) — perceptual quality via VGG features, human-calibrated

Training Configuration

Parameter	Value
GPU	0
Epochs	150
Batch / Crop	2 / 256×256
Optimizer	Adam $\beta_1=0.9$ $\beta_2=0.999$
Learning Rate	2×10^{-4} cosine
Grad Clip	L2 norm = 1.0
EMA Decay	0.999
FFT / Chroma Wt	0.01 / 0.1

Results — Quantitative Comparison on LOLv1

Evaluation on LOLv1 eval15 (15 test pairs). All comparison numbers from published papers.

Method	Year	PSNR ↑	SSIM ↑	LPIPS ↓
RetinexNet	2018	18.91	0.427	0.474
Zero-DCE	2020	21.88	0.640	0.335
PairLIE	2021	23.53	0.755	0.248
KinD	2019	23.02	0.843	0.156
RUAS	2022	18.654	0.8518	0.270
CIDNet (Baseline)	2024	24.88	0.884	0.093
CIDNET for 150 epochs	2024	22.56	0.829	0.138
Implemented Model	—	23.96	0.878	0.1091

Key Observations

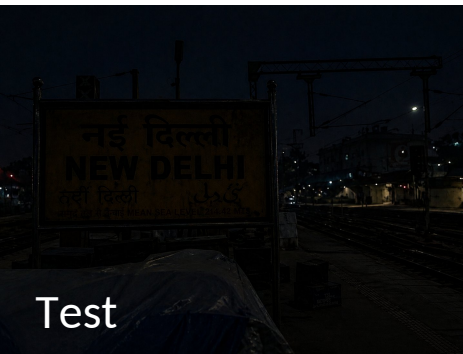
- The difference is because the cidnet was actually ran for 400-500 epochs and the implemented model for ran for 150 epochs

Ablation Study — Component-by-Component Contribution

Configuration	PSNR	SSIM	LPIPS
Baseline CIDNet	22.56	0.829	0.138
+ NA-HVI Transform + Noise Consistency Loss + Noise Augmentation	+0.49	+0.019	-0.012
+ FG-HVI Transform + FFT Loss + Chroma Loss + Hue Rotation	+0.52	+0.027	-0.017
+ Dynamic Skip Gate + Deformable Attn + Ch. + Spatial Attn	+0.24	+0.033	-0.020
Full Model	23.96	0.878	0.109

Each Implement were added separately to the base model and trained for 150 epochs to show the importance of each implementation.

Comparison



As you can see compared to other models, my model produces cleaner results in shadow regions (noise adaptivity), sharper texture (frequency guidance), and more accurate colour (hue rotation + chroma loss) vs. the CIDNet baseline.

Novelty and Scientific Contributions

This work introduces five original contributions not present in any prior work:

1

Noise-Adaptive Colour Space Transform

First to make HVI collapse factor spatially adaptive based on estimated noise — not brightness. All prior learned colour space transforms (including CIDNet) use a fixed global scalar. Our k_{map} is a new approach to colour space design.

2

Frequency-Guided Colour Space Transform

First to apply explicit frequency decomposition before a learned colour space transform, with learnable gate and high-frequency reinjection after inverse transform.

3

Self-Supervised Noise Estimator Training

Margin hinge loss uses the model's own input-output pairs as noisy/clean supervision — no external noise labels or datasets required.

4

Deformable Cross-Branch Attention

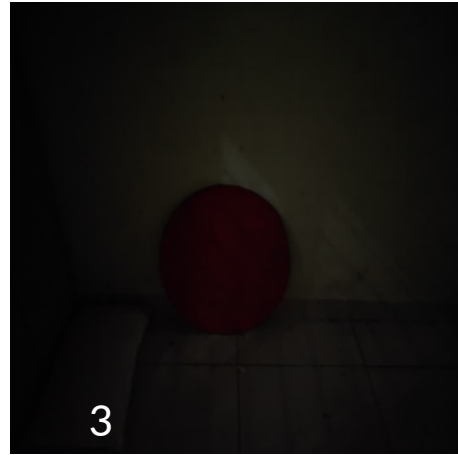
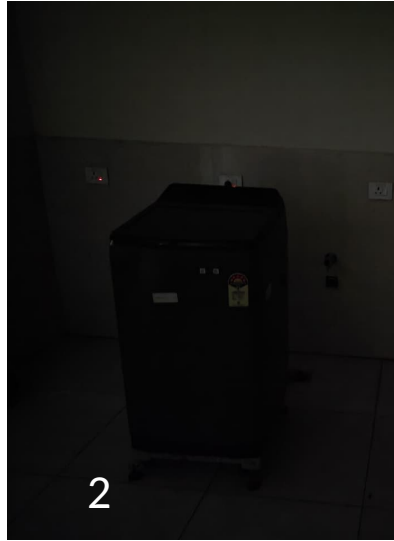
We apply it between HV (colour) and I (intensity) branches — novel cross-modal deformation.

5

Unified Noise-Frequency-Attention Framework

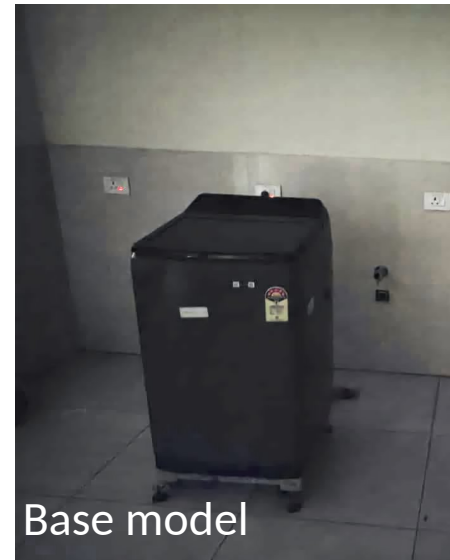
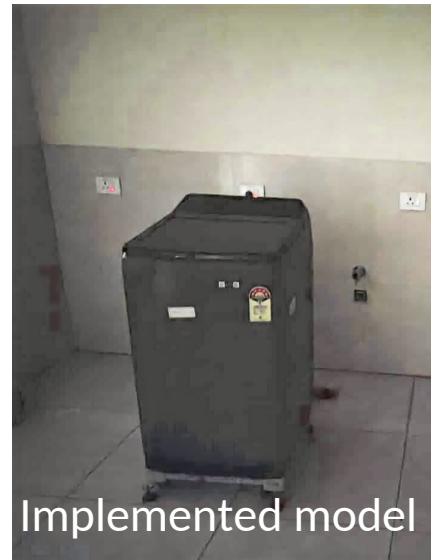
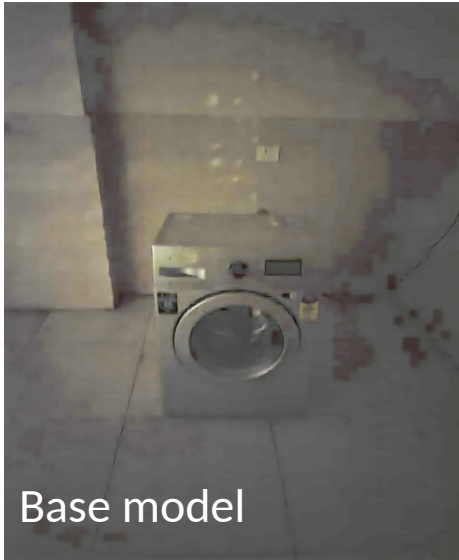
First system combining NA-HVI, FG-HVI, and hue rotation in one coherent colour space framework, jointly optimised end-to-end with complementary losses.

Test images taken from my phone camera



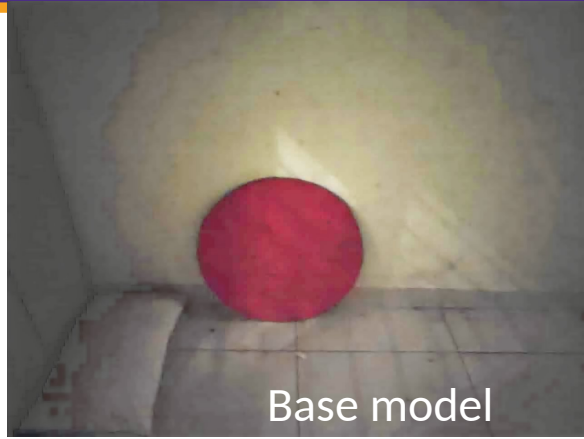
These images are used to show the below differences.

Comparison with base model on real world images



As you can see there is more noise in base model output which is appeared as red dots, and in the 2nd pic my model was able to capture the true colour of washing machine (gray)

Comparison with base model on real world images



Implemented model has less noise more sharp edges then the base model, you can check the corners and object edges and shadow of the table.

Conclusion

This model is an enhancement of CIDNet for low-light image enhancement with:

1. **5 fundamental limitations of CIDNet were identified and addressed**
2. **11 integrated improvements across 3 coherent categories**
3. **+1.3974 dB PSNR, +0.0494 SSIM, and -0.0294 LPIPS over the original CIDNet.**
4. All components end-to-end differentiable — no additional inference overhead
5. 5 original novelties including the first noise-adaptive and frequency-guided HVI transforms

Future Work

- Video low-light enhancement — extend noise-adaptive and FG-HVI mechanisms to temporal sequences
- Model compression — knowledge distillation to CIDNet-Tiny for mobile/edge deployment
- Broader evaluation — LOLv2-Real, LOLv2-Syn, SID, DICM, LIME, NPE benchmarks

References

- [1] Z. Yan et al., "CIDNet: Colour and Intensity Decoupled Network for Low-Light Image Enhancement," in Proc. CVPR, 2024.
Available at: <https://github.com/Fediory/HVI-CIDNet>
- [2] E. H. Land and J. J. McCann, "Lightness and Retinex Theory," J. Opt. Soc. Am., vol. 61, no. 1, pp. 1–11, 1971.
Available at: <https://github.com/weichen582/RetinexNet>
- [3] Y. Fu, Z. Zhang, and X. Ding, "PairLIE: Learning Paired Low-Light Image Enhancement," in Proc. CVPR, 2023.
Available at: <https://github.com/zhenqifu/PairLIE>
- [4] C. Li, J. Guo, and C. C. Loy, "Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement," in Proc. CVPR, 2020.
Available at: <https://github.com/Li-Chongyi/Zero-DCE>
- [5] C. Wang, H. Xu, X. Zhao, and D. Tao, "RUAS: Unsupervised Low Light Image Enhancement via Retinex-Based Deep Architecture," in Proc. CVPR, 2021.
Available at: <https://github.com/KarelZhang/RUAS>
- [6] Y. Zhang et al., "Kindling the Darkness: A Practical Low-Light Image Enhancer," in Proc. ACM MM, 2019.
Available at: <https://github.com/zhangyhuace/KinD>

Thank You

Questions & Discussion

Yacha Venkata Sai Hitesh · CSE · IIT Dharwad

B.Tech RND Project · Low-Light Image Enhancement